

Key Recovery Attack on FEAT_PACQARMA3 Feature

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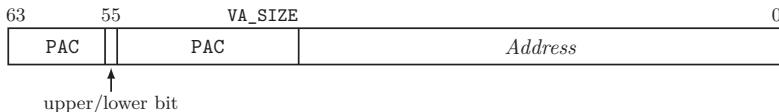
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Arm's FEAT_PACQARMA3 Feature and Summary of the Target

The Pointer Authentication Code (PAC) feature

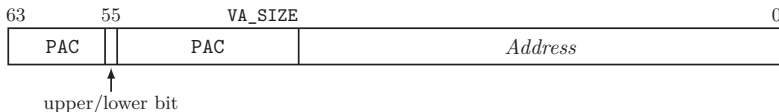
1. Introduced in [Arm V8](#) architecture in 2016 [[ARM16](#)] for enforcing CFI
 - 1.1 Tag $\leftarrow$ **PRF**(Return Address, Pointer's Context Information)
 - 1.2 Inserts tag on function entry in reserved bits of pointers and verifies it upon exit



2. Discard some output bits of a **lightweight** Tweakable Block Cipher (TBC)
3. Protects against attacks targeting stack manipulation
4. **First line of defense** to system software and complex programs, e.g., Browsers

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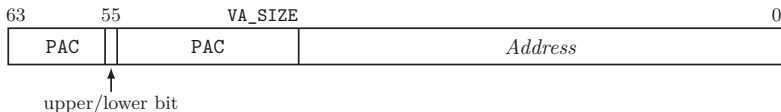
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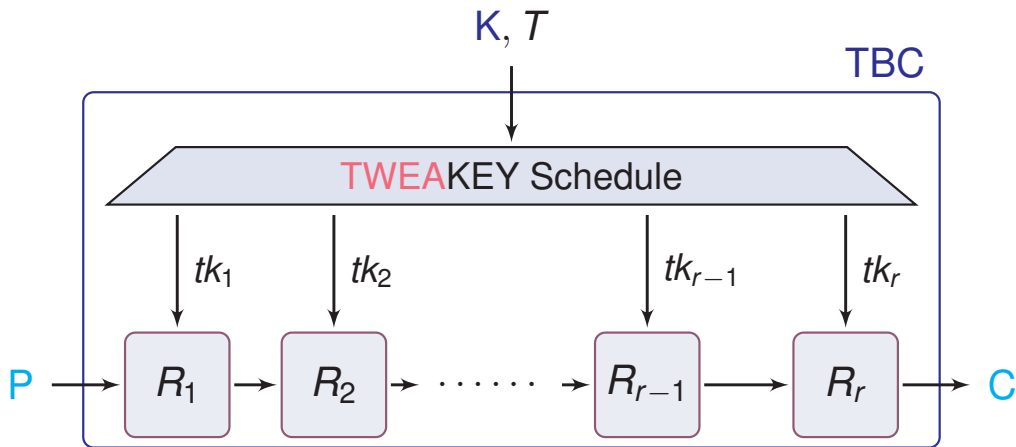
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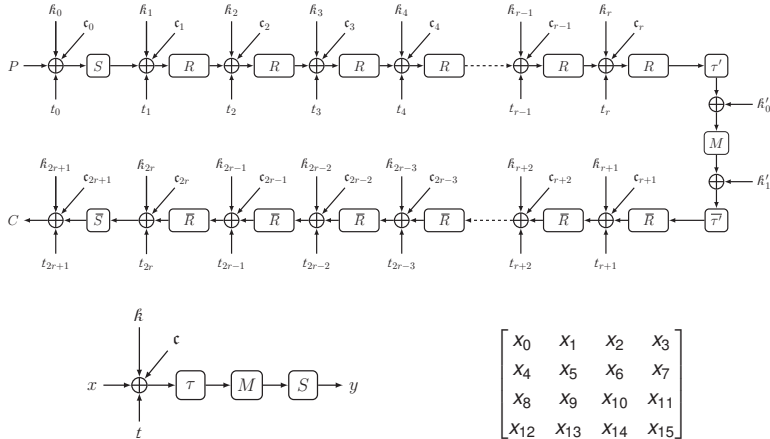


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Iterated Tweakable Block Cipher with TWEAKEY Schedule



QARMA_{v1} [Ava17]: High-Level View



The S Layer and The M Layer

$$\begin{bmatrix} x_0 & x_1 & x_2 & x_3 \\ x_4 & x_5 & x_6 & x_7 \\ x_8 & x_9 & x_{10} & x_{11} \\ x_{12} & x_{13} & x_{14} & x_{15} \end{bmatrix} \xrightarrow{\text{S Layer}} \begin{bmatrix} s(x_0) & s(x_1) & s(x_2) & s(x_3) \\ s(x_4) & s(x_5) & s(x_6) & s(x_7) \\ s(x_8) & s(x_9) & s(x_{10}) & s(x_{11}) \\ s(x_{12}) & s(x_{13}) & s(x_{14}) & s(x_{15}) \end{bmatrix}, S: \mathbb{F}_2^4 \rightarrow \mathbb{F}_2^4$$

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Example: $(x_4 \lll a) \oplus (x_8 \lll b) \oplus (x_{12} \lll c)$

$M = M^{-1}$ over the ring $R_4 = \mathbb{F}_2[X]/(X^4 + 1)$

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State Mapped by τ to Different Columns

0			
	5		
		10	
			15

 $\xrightarrow{\tau}$

0			
10			
5			
15			

	1		
4			
			11
		14	

 $\xrightarrow{\tau}$

11			
1			
14			
4			

			3
		6	
	9		
12			

 $\xrightarrow{\tau}$

		6	
		12	
		3	
		9	

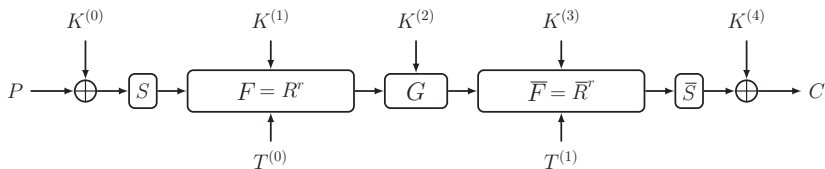
		2	
			7
8			
	13		

 $\xrightarrow{\tau}$

			13
			7
			8
			2

QARMA_{v1} [Ava17]: The Tweakable Block Cipher

1. QARMA_{v1} is a **reflection cipher**: $E = F^{-1} \circ G \circ F : \mathbb{F}_2^{64} \rightarrow \mathbb{F}_2^{64}$

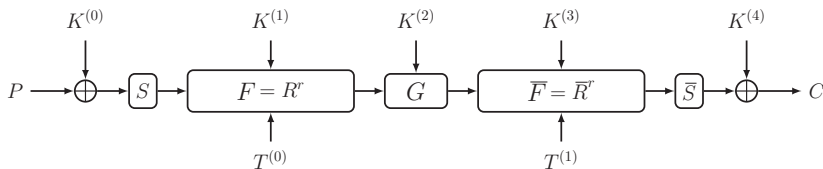


2. QARMA_{v1} as the PAC feature:

- 2.1 Pointer is used as the **"clamped"** plaintext input ($64 - \text{VA}$ bits are set to 0)
- 2.2 Context is used as the tweak
- 2.3 The tag is the **"chopped"** ciphertext (attacker can only observe $64 - \text{VA}$ bits)

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Summary of The Target

1. FEAT_PACQARMA5 uses QARMA₅, the 12-round version of QARMA_{v1} (100 GE)
2. FEAT_PACQARMA3 feature uses QARMA₃, the 8-round TBC QARMA_{v1} (66 GE)
3. Cryptanalysis of such a heavily round-reduced cipher is important
4. KPA: memory read gadgets are common, adversary can read signed pointers
5. CPA: JIT environments (e.g., JavaScript), allow to generate chosen pointer
6. Challenges in the cryptanalysis of FEAT_PACQARMA3:
 - 6.1 Clamping and chopping limits the differential characteristics we can use
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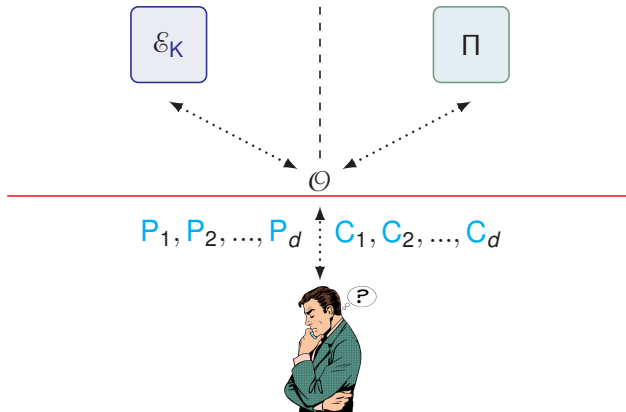
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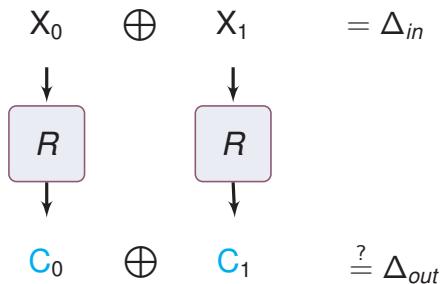
Differential Cryptanalysis of FEAT_PACQARMA3

Cryptanalysis 1: Real vs Ideal Distinguishing Game

$\mathcal{E}: \mathcal{P} \times \mathcal{K} \rightarrow \mathcal{C}$ is a block cipher and Π is a random permutation

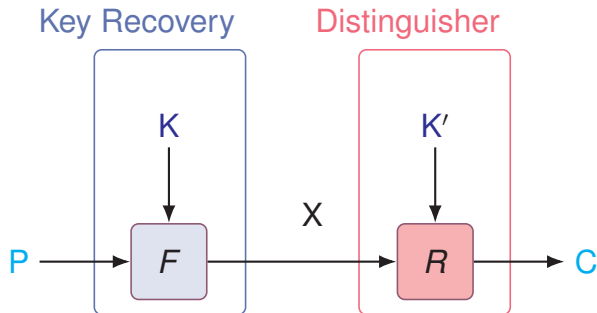


Differential Distinguisher



$$\Pr \left(R(X) \oplus R(X \oplus \Delta_{in}) = \Delta_{out} \right) = 2^{-q}, q \ll n$$

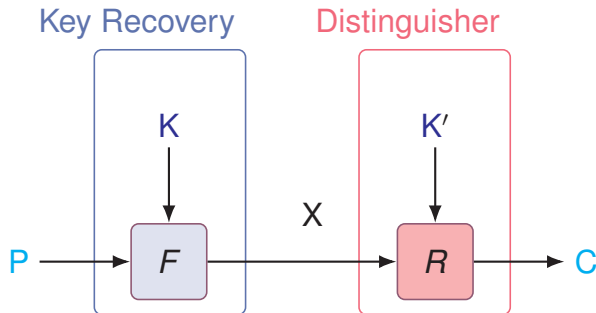
Key Recovery Using Differential Distinguisher



► $R(\cdot) = R_{r-1} \circ R_{r-2} \circ \dots \circ R_1 \circ R_i(\cdot)$

► $F(P_0, K) \oplus F(P_1, K) = X_0 \oplus X_1 = \Delta_{in} \xrightarrow{2^{-q}} \Delta_{out}$, for the correct key K

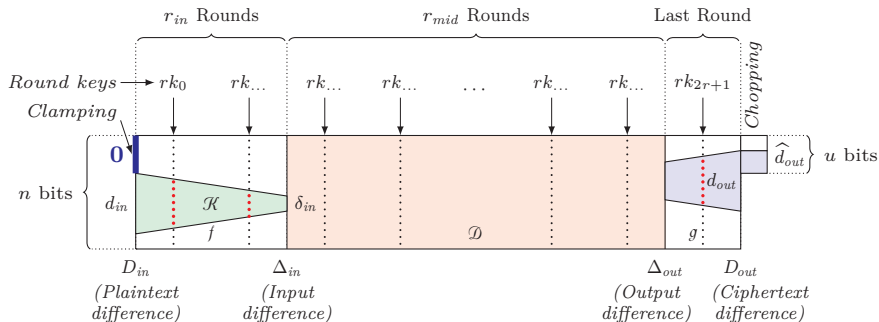
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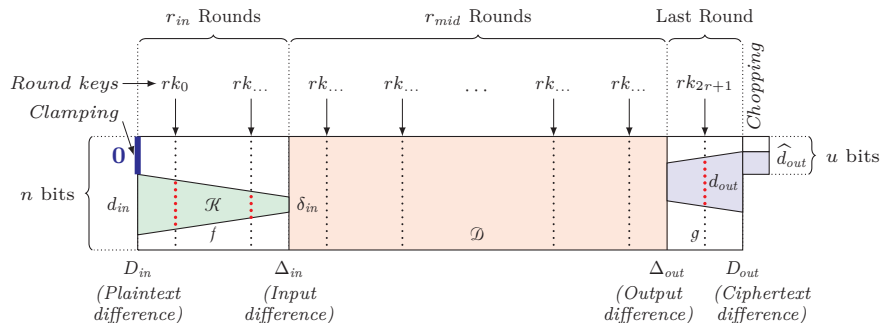
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Differential Attack on a Clamped and Chopped Cipher



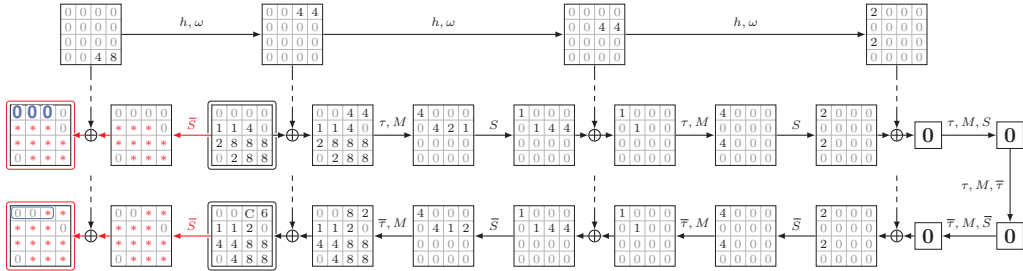
- ▶ No difference, and no control over the **clamped** bits
- ▶ Output is **chopped**, the adversary can observe only u bits
- ▶ $q \triangleq -\log_2(\Pr[\Delta_{in} \rightarrow \Delta_{out}])$

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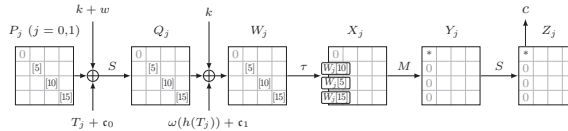
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An Example of $VA = 52$

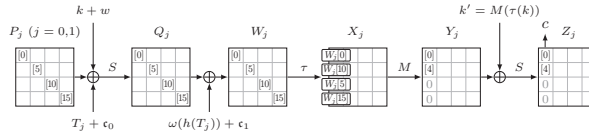


$$d_{in} = 4 \times 10 = 40, u = 12, \hat{d}_{out} = 4, q = 12, \text{ i.e., } \Pr[\Delta_{in} \rightarrow \Delta_{out}] = 2^{-12}$$

Key Recovery Goal



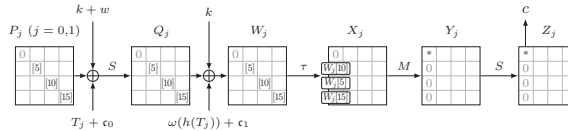
The 128-bit master key is $K = w || k$ and let, $k' = M(\tau(k))$



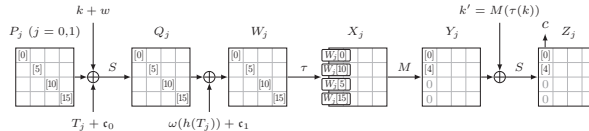
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Time: $\leq 2^{64}$ (Break) and $\leq 2^{80}$ (Warning)

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Overview of the Attacks

Structures

$\mathcal{S} : \mathcal{S}_0, \mathcal{S}_1 \subseteq \mathbb{F}_2^n \times \mathbb{F}_2^{n_t}$ and $\mathcal{S}_1 = (0, \mathbf{t}_0) + \mathcal{S}_0$. $|\mathcal{S}_i| = 2^{d_{in}}$, $|\mathcal{S}| = 2^{2d_{in}}$ with data complexity $2^{d_{in}+1}$

Pair Sieving

```
1: for  $((P_0, T_0), (P_1, T_1)) \in \mathcal{S}_0 \times \mathcal{S}_1$  do
2:   if  $\varepsilon_{T_0, K}(P_0) + \varepsilon_{T_1, K}(P_1) \in D_{out}$  then  $\mathcal{C} = \mathcal{C} \cup ((P_0, T_0, C_0), (P_1, T_1, C_1))$ 
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Guess-and-Filter

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1: Prepare a dictionary  $\chi$  associated with the elements  $k \in \langle \mathcal{K} \rangle$  (AVL trees)
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4:     if  $f_k(P_0^i, T_0^i) + f_k(P_1^i, T_1^i) \in \Delta_{in}$  then  $\chi(k) = \chi(k) + 1$ 
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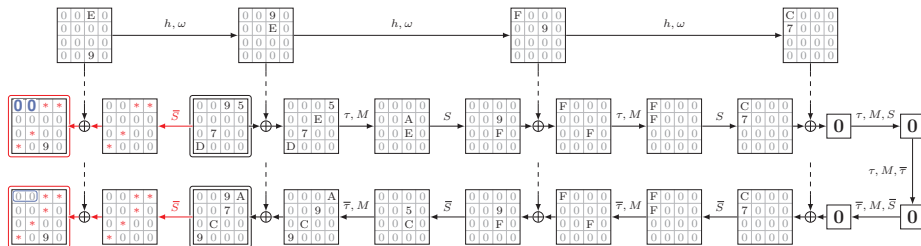
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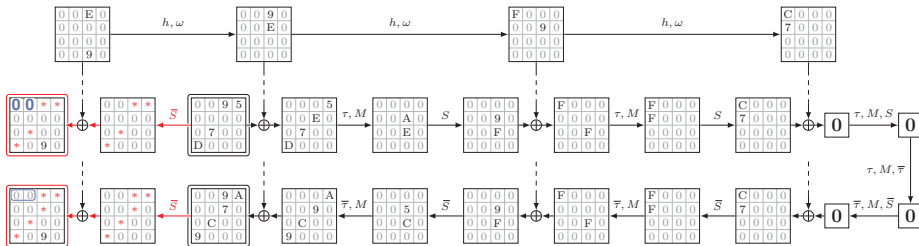
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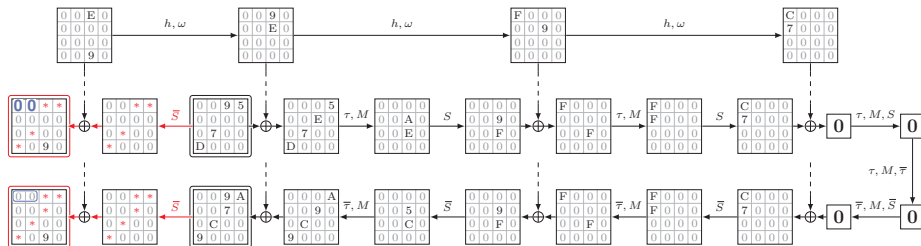


Attack on $V_A = 56$ (Char. 1)



0.5R Attack: Recover the 16 key bits $(k + w)[\{2, 3, 9, 12\}]$

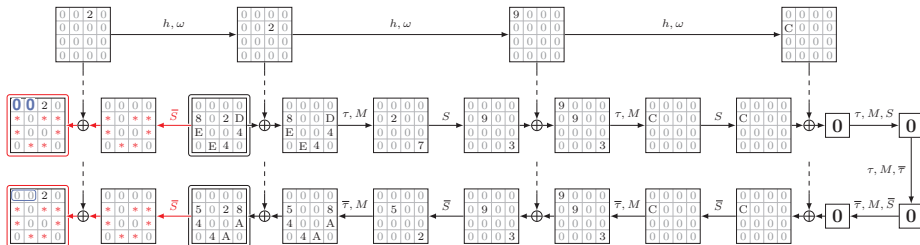
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Peeling: Recover the 8 key bits $k'[\{6, 10\}]$ using $(k + w)[\{6\}]$

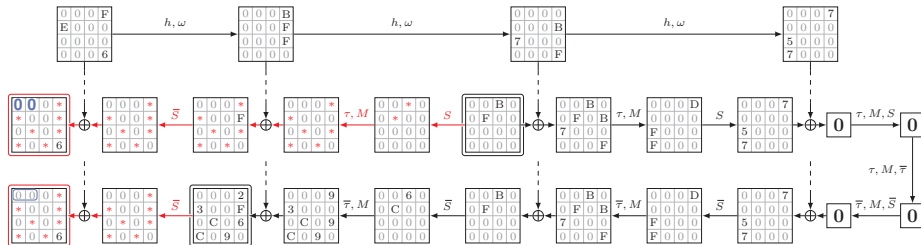
Attack on $VA = 56$ (Char. 2)



0.5R Attack: Recover the 28 key bits $(k + w)[\{4, 6, 7, 8, 11, 13, 14\}]$

Peeling: Recover the 8 key bits $k'[\{5, 15\}]$ by guessing $(k + w)[\{1\}]$

Attack on VA = 56 (Char. 3)



Peeling: Recover the 4 key bits $k'[2]$ using $(k + w)[\{3, 9, 12\}]$

Summarized Attack Results on $VA = 56$

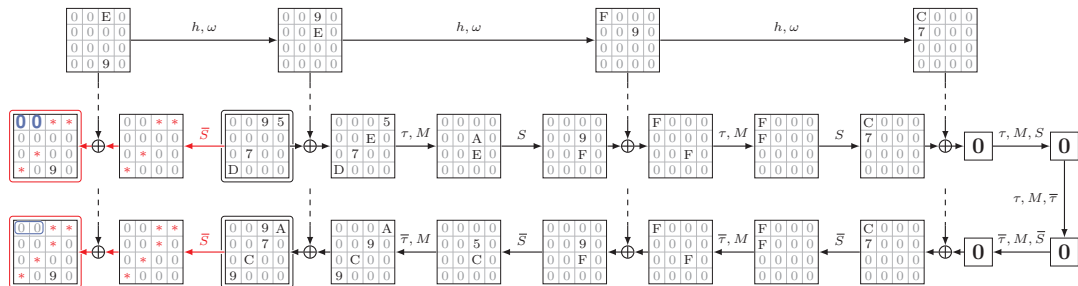
$$(k + w) \leftarrow \begin{bmatrix} b & 1 & 1 & 2 \\ 1 & b & 1 & 1 \\ 1 & 2 & b & 1 \\ 2 & 1 & 1 & b \end{bmatrix} \quad \text{and} \quad k' \leftarrow \begin{bmatrix} b & b & 3 & b \\ b & 1 & 2 & b \\ b & b & 2 & b \\ b & b & b & 1 \end{bmatrix}$$

Time Complexity $\approx 2^{60} T_{\mathcal{E}}$ (dominated by the brute force step)

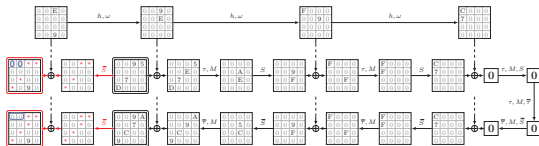
Data Complexity $\approx 2^{25}$ Chosen Plaintexts

Data Complexity, Time Complexity and Success Probability

An Example of 0.5R Attack on $VA = 56$



An Example of 0.5R Attack on $V_A = 56$



1. We have $q = 10$, $d_{in} = 16$
2. 16 key bits $(k + w)[\{2, 3, 9, 12\}]$
3. $u - \hat{d}_{out} = 8$

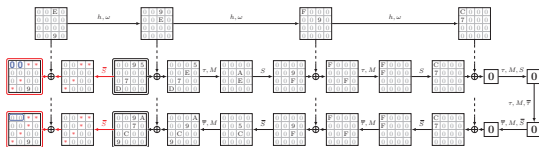
N : #pairs satisfying $\Delta_{in}??$

R : Counters Corr. to "Right keys"

W : Counters Corr. to "Wrong keys"

1. $W \sim \text{Poisson}(N \cdot 2^{-8})$
2. $R \sim \text{Poisson}(N \cdot (2^{-8} + 2^{-10}))$
3. $\sigma = \sqrt{N \cdot 2^{-8} (1 - 2^{-8})}$

An Example of 0.5R Attack on $V_A = 56$



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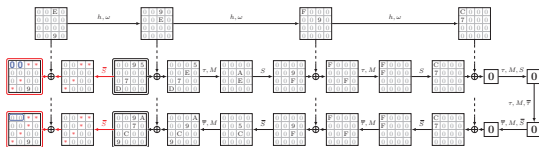
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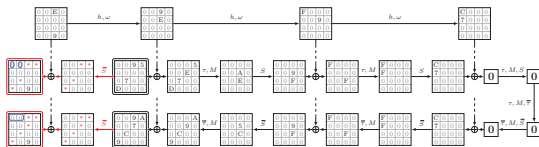
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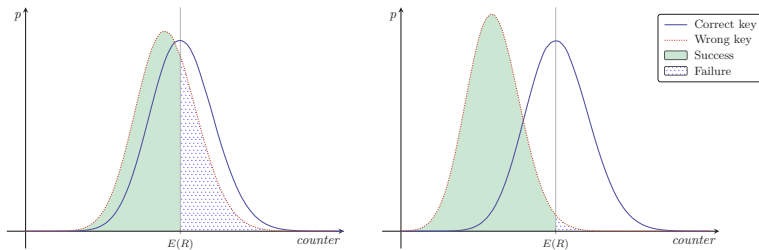
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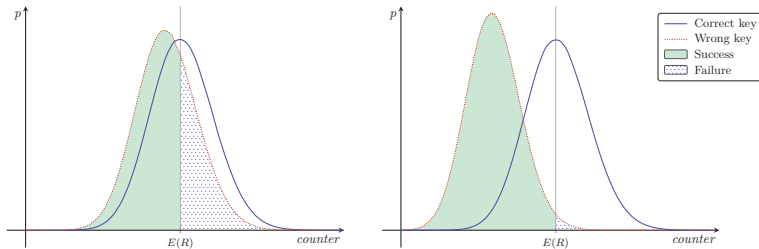
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Counters Corresponding to Right vs Wrong Keys



1. $E[R] - E[W] \geq \gamma\sigma \implies N \cdot (2^{-q} + 2^{-u+\hat{d}_{out}}) - N \cdot (2^{-u+\hat{d}_{out}}) \geq \gamma \sqrt{N \cdot 2^{-u+\hat{d}_{out}} \cdot (1 - 2^{-u+\hat{d}_{out}})}$
2. Success Probability: $\left(\sum_{a \leq E(R)} \Pr[W = a] \right)^{\nu-1}$, where ν is the #possible keys

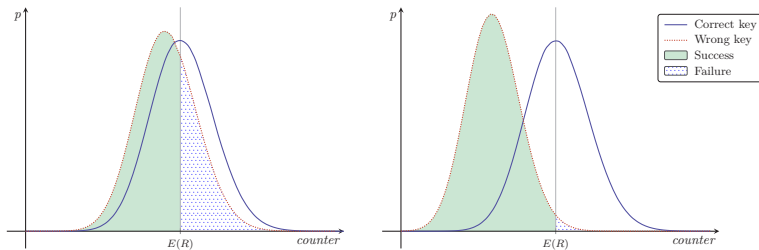
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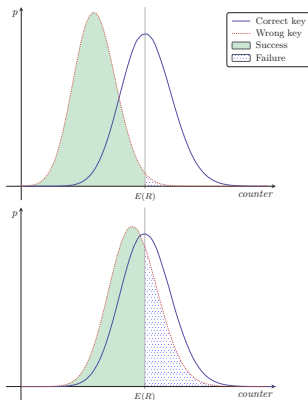
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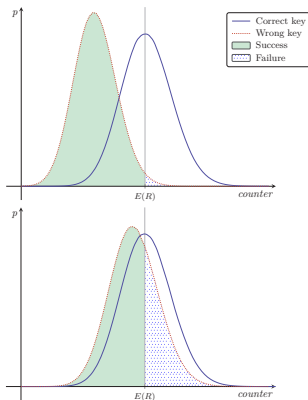
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4. $\gamma = 8$ gives $N \approx 2^{18}$
5. So we need 2^2 structures of size 2^{16}
6. Data Complexity: $2 \times 2^2 \times 2^{16} = 2^{19}$
7. Sieved Pairs: $L \approx 2^{26}$
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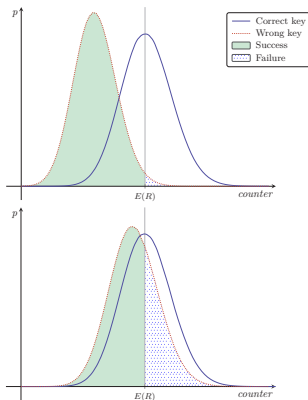
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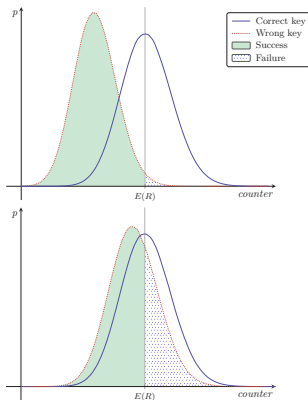
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Experimental Verification

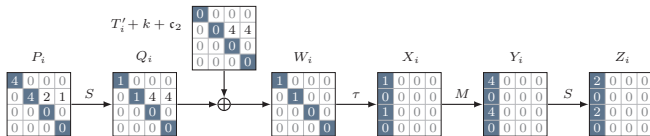
Experimental Verification

1. $\bar{h}_i$: # pairs that satisfy D_{out} in the i -th experiment and random noise $\eta = N \cdot 2^{-u+\hat{d}_{out}}$
2. $\bar{q} = -\log_2(\text{Mean}(\bar{h}_i - \eta)/N)$
3. If $\bar{h}_i - \eta > (1/2)\gamma\sigma$, we consider the experiment to have been successful

Char.	$\langle u - \hat{d}_{out} \rangle$	N	σ	$\bar{h}_i - \eta$			q	$\bar{q}$	Successes /100 trials
				Mean	$\pm$	S.D.			
1	8	2^{14}	7.98	64.11	$\pm$	11.69	7.98	8.00	100
2	8	2^{18}	31.94	256.19	$\pm$	32.40	10.01	10.00	100
3	12	2^{18}	7.99	63.80	$\pm$	11.50	12.01	12.00	100
4	8	2^{22}	127.75	1022.33	$\pm$	1832.27	12.00	12.00	27
4*	8	2^{18}	31.93	258.32	$\pm$	39.27	10.00	9.98	100
5	12	2^{29}	361.99	2128.71	$\pm$	2075.29	18.00	17.94	53
5*	12	2^{27}	181.00	1026.25	$\pm$	174.94	17.00	17.00	100
6	16	2^{31}	181.02	1028.42	$\pm$	176.56	21.00	20.99	100
7	4	2^{33}	5781.29	33324.88	$\pm$	6337.25	18.00	17.97	100



Conditional Characteristic (Char. 4)



$$\Delta_P \rightarrow \Delta_Q \Rightarrow \mathcal{A} = \{x : S(x+4) + S(x) = 1\} = \{1, 3, 5, 7\}$$

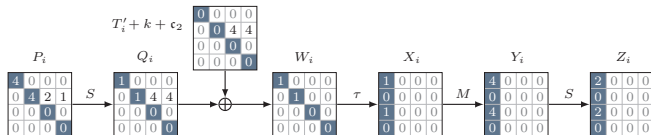
$$\Delta_Y \rightarrow \Delta_Z \Rightarrow \mathcal{B} = \{x : S(x+4) + S(x) = 2\} = \{10, 11, 14, 15\}$$

$$(T'_i + k + c_2)[5] + (T'_i + k + c_2)[0] \in \{0, 1, 4, 5\}$$

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AutoDiVer: Automatically Verifying Differential Characteristics and Learning Key Conditions. IACR Trans. Symmetric Cryptol., 2025(1):471–514, 2025

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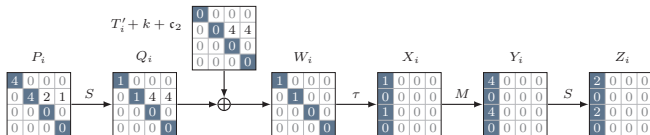
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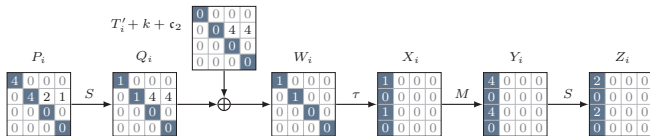
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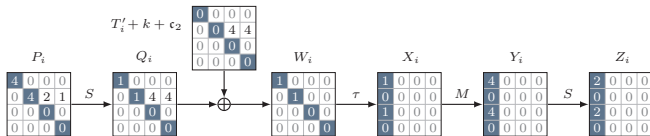
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Conditions. IACR Trans. Symmetric Cryptol., 2025(1):471–514, 2025

Results

Results

Rounds attacked	VA & Profile	Attack Complexity			Technique
		Data	Time	Memory	
4 + 4	56 (A)	2^{25} CP	2^{60}	$2^{28.98}$	RT Diff.
4 + 4	52 (A)	$2^{25.3}$ CP	2^{55}	$2^{28.94}$	RT Diff.
4 + 4	48 (A)	2^{28} CP	2^{60}	$2^{40.83}$	RT Mult. Diff.
4 + 4	44 (A)	$2^{31.19}$ CP	2^{68}	$2^{31.67}$	RT Diff.
4 + 4	40 (A)	$2^{37.5}$ CP	2^{76}	$2^{38.82}$	RT Diff.
4 + 4	36 (A)	$2^{33.7}$ CP	2^{76}	$2^{35.02}$	RT Diff.
4 + 4	32 (A)	2^{42} CP	2^{76}	$2^{44.32}$	RT Diff.
4 + 4	32 (M)	2^{45} CP	2^{76}	$2^{46.32}$	RT Diff.

*RT: Related-Tweak

Conclusion

1. Practical attacks on QARMA₃ used in FEAT_PACQARMA3
 - 1.1 Despite (clamping, chopping), our attacks outperform previous ones on QARMA₃
 - 1.2 Versions with $\text{VA} = 44, 48, 52$, and 56 do not meet the expected security level
2. Finding good characteristics is challenging in clamped and chopped ciphers
 - 2.1 We designed a bit-wise model of the r_{mid} middle rounds as a program for STP
 - 2.2 Maximizing the prob. of a characteristic (2^{-p}) alone is insufficient $\rightarrow p + d_{\text{in}}$
 - 2.3 We observed strong clustering effects.
3. Future directions
 - 3.1 Our results do not directly translate to practical breaks of the *Pointer Authentication Code (PAC) feature*
 - 3.2 Investigating practical constraints that hinder real-world attacks
 - 3.3 Extending our framework to analyze the security of QARMAv2 within PAC *feature*

Conclusion

1. Practical attacks on QARMA_3 used in FEAT_PACQARMA3
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Conclusion

1. Practical attacks on QARMA_3 used in FEAT_PACQARMA_3
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Thank You for your
attention!

Any questions?



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ARM Connected blog. ARMv8-A architecture — 2016 additions.

<https://www.community.arm.com/processors/b/blog/posts/armv8-a-architecture-2016-additions>, **October 2016.**



Roberto Avanzi.

The QARMA Block Cipher Family. Almost MDS Matrices Over Rings With Zero Divisors, Nearly Symmetric Even-Mansour Constructions With Non-Involutory Central Rounds, and Search Heuristics for Low-Latency S-Boxes.

IACR Trans. Symmetric Cryptol., 2017(1):4–44, 2017.